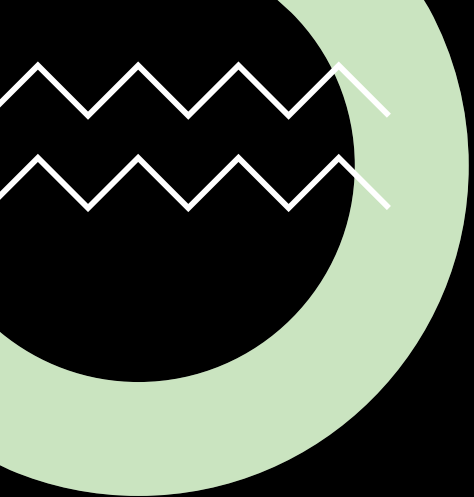




Rethinking Statistical Quality in a Data-Rich World

Deming Lecture

August 4, 2026

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Rethinking Statistical Quality in a Data-Rich World

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Any opinions or viewpoints shared today are those of the author, Sallie Keller, and may not reflect those of the U.S. Census Bureau.



"Quality should be built into the process, not inspected in at the end."

— W. Edwards Deming

For decades, official statistics have relied on well-established principles for ensuring quality. Those principles have served us extraordinarily well.

But the world in which they were developed has changed.



Deming gave us a timeless principle.

Our challenge is to apply that principle in a fundamentally different statistical environment.

The World Has Changed



What Changed?



declining response rates



administrative records



linked data



AI



rising expectations

The Last Statistical Mile

How do we bridge development, production and use?





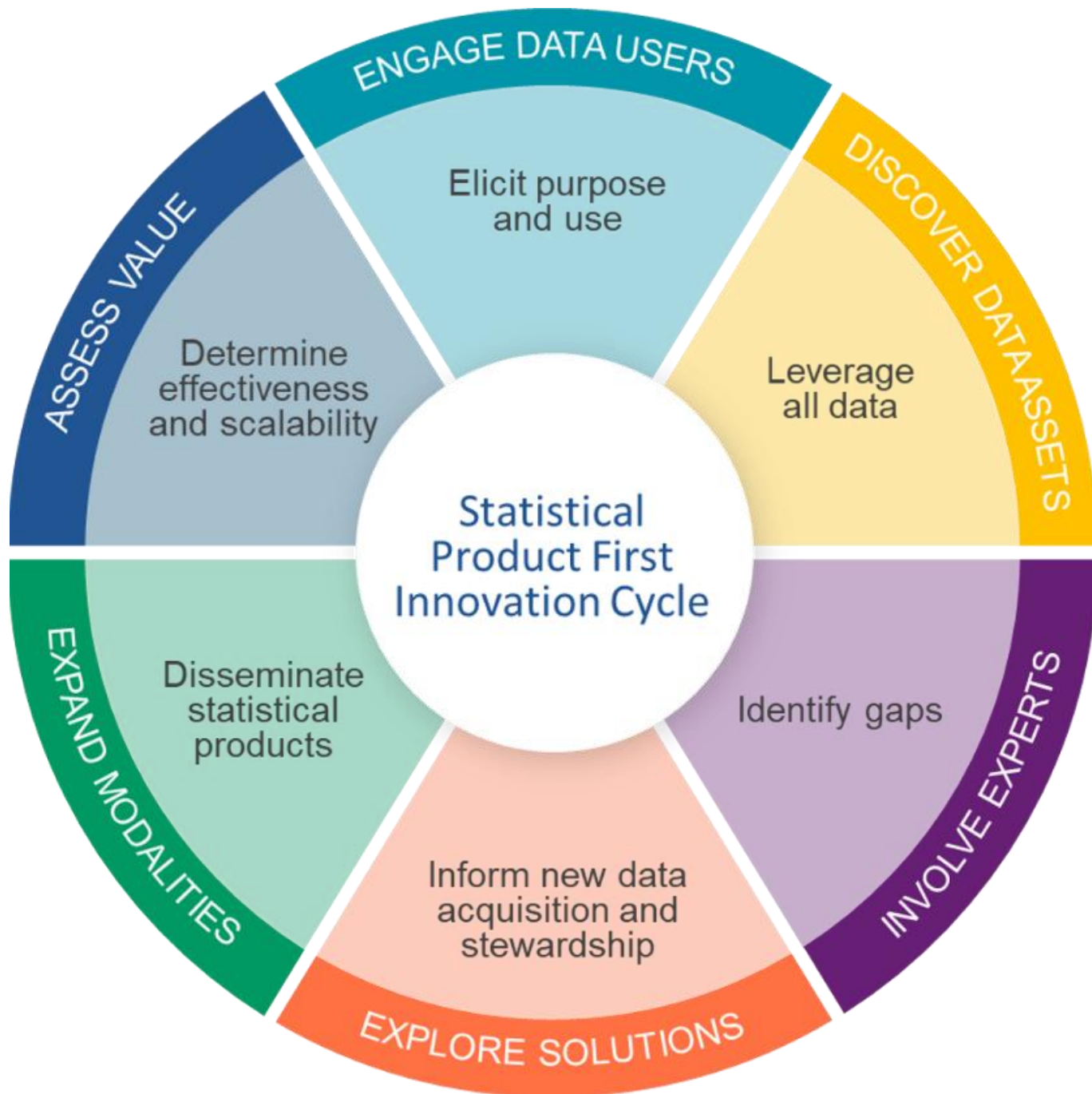
We Keep Asking the Wrong Question

Instead of asking:

*“What data do you
need?”*

We should be asking:

*“What decision are we
trying to support?”*

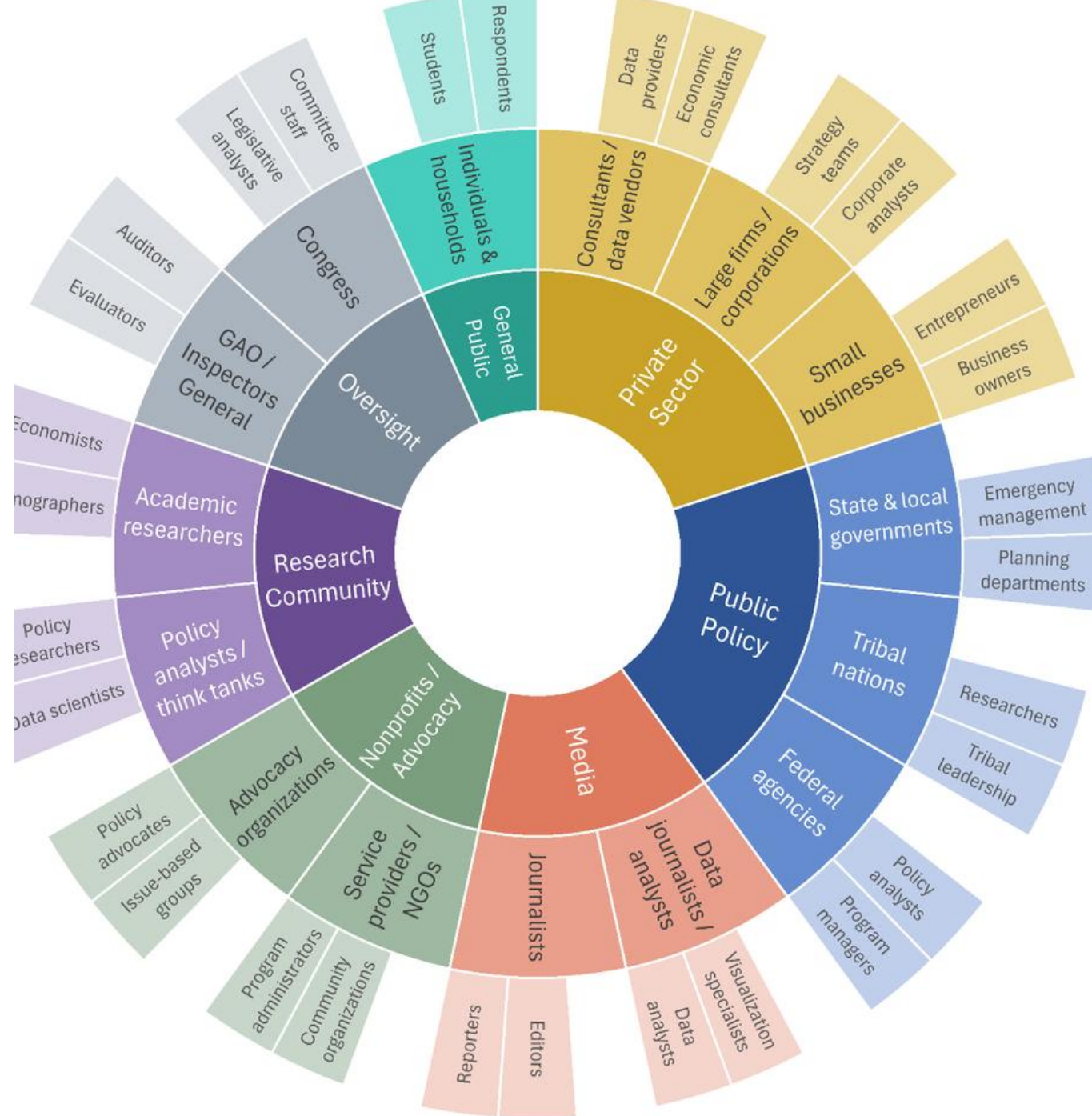


Statistical Product First

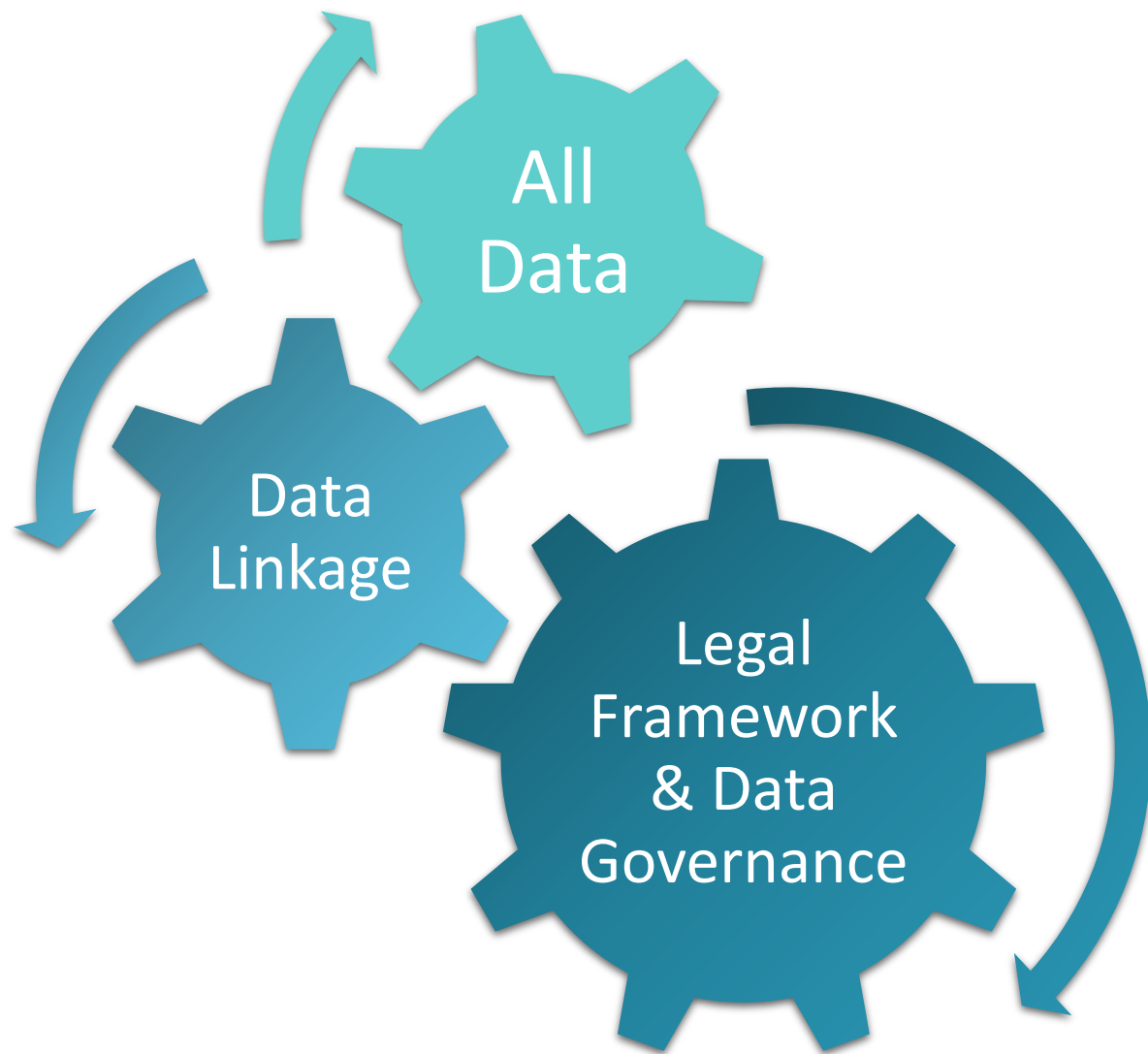
Know Your Users



Segmenting Stakeholders through a Hierarchical Approach





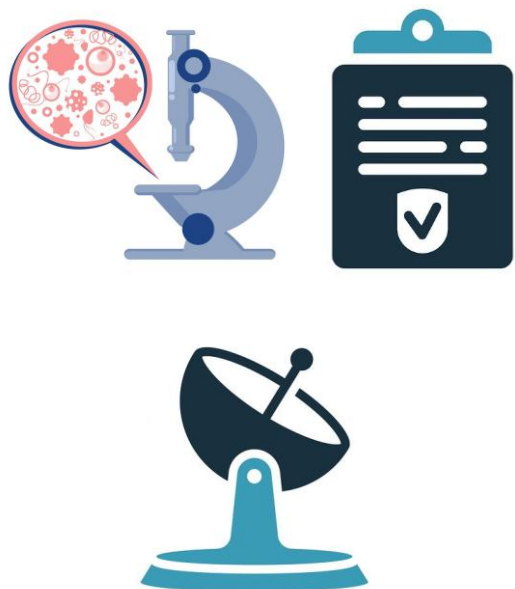


Leverage Data Infrastructure

Use *ALL* Data Assets

Going beyond the survey data we collect

Designed Data



Administrative Data



Opportunity Data



Procedural Data



Harness Data Linkage

We collect a variety of linkage fields and tailor linkage methodology to each linkage type – people, places, jobs, and organizations.

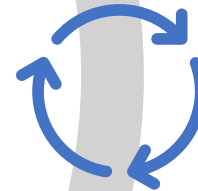


Statistical Product Spectrum for *ALL* Users

An opportunity
to identify and fill
the space in between



Protected microdata
in Federal Statistical
Research Data Centers
(FSRDCs)



Simple, public
use products

Adhere to Legal Framework and Data Governance

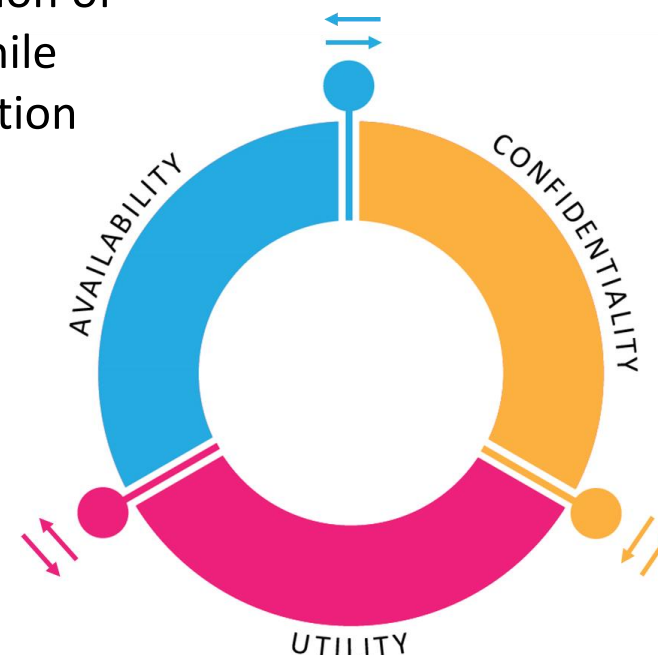
***Title 13** directs us to acquire and use external data records for statistical purposes.*

***Foundations for Evidence-Based Policymaking Act** promotes and encourages data sharing.*

***Confidential Information Protection and Statistical Efficiency Act** sets forth functional separation of statistical versus administrative uses of data.*

Ethical Data Stewardship

Requires confidential protection of data subjects' information while exclusively using this information for statistical purposes.





What does it look like when we
put this into practice?

Reenvisioning Community-level Statistics

Vision

To produce community-level statistics using all relevant data.

Mission

To develop a unified multisource data framework that leverages the latest advances in science and technology to produce high-quality, relevant, timely, and granular statistics with measures of uncertainty to meet stakeholder needs.

Insights to Achieve Success

01

Survey as core input

Novel approaches with all available, relevant data will generate new value and help mitigate some program risks

02

Ad Rec as a mode

Treat administrative records as an input mode, not only a Band-Aid or enhancement

03

Put modeling to work

Utilize to unlock the potential of all data sources and provide blended statistics at level of granularity and timeliness that stakeholders need



Identifying Risk of Homelessness for Veterans

Advancing proactive decision-making through the Statistical Product First (SPF) Innovation Cycle

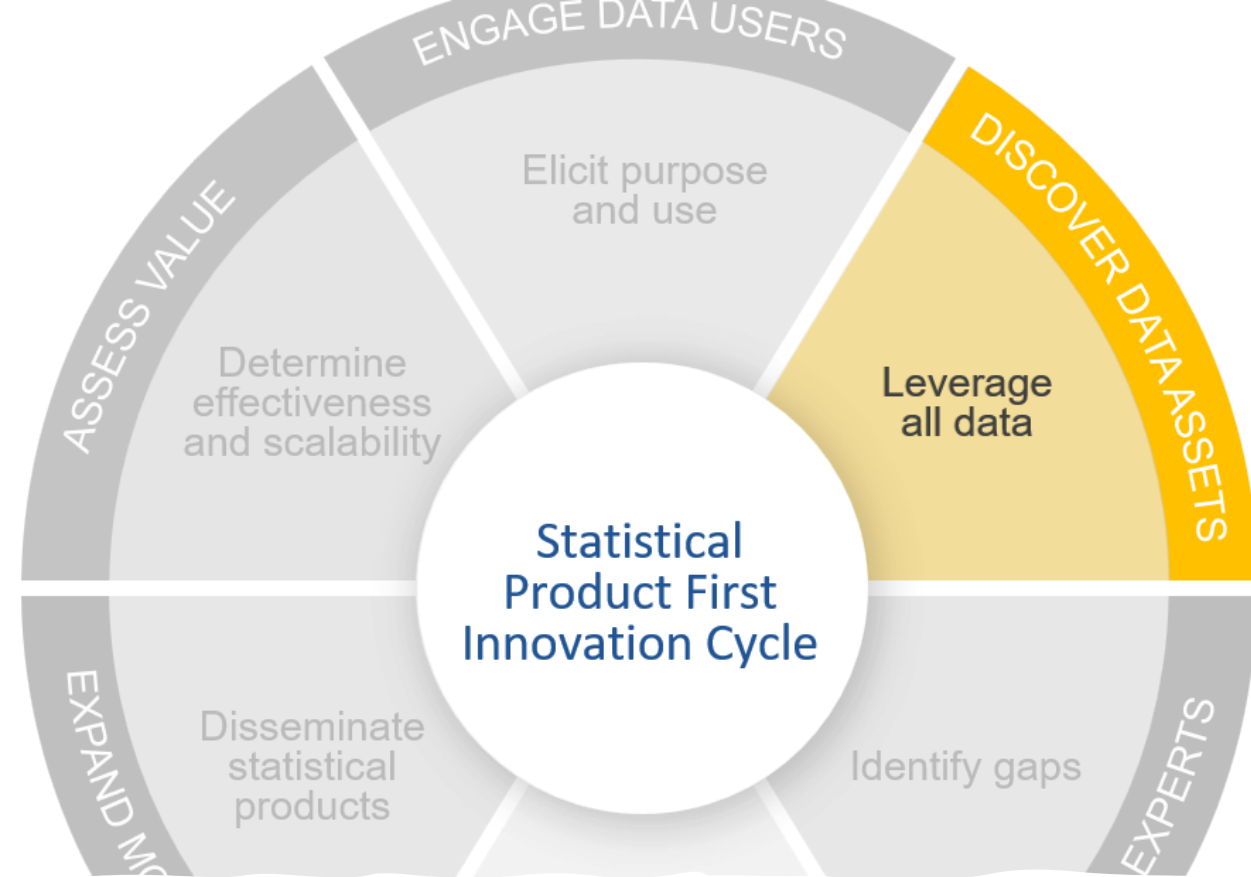


Engage Users

Elicit Purpose and Use

What decision are we trying to support?

- Identify veterans at elevated risk **before** homelessness occurs
- Support earlier, more targeted intervention
- Design statistics around stakeholder decisions—not datasets



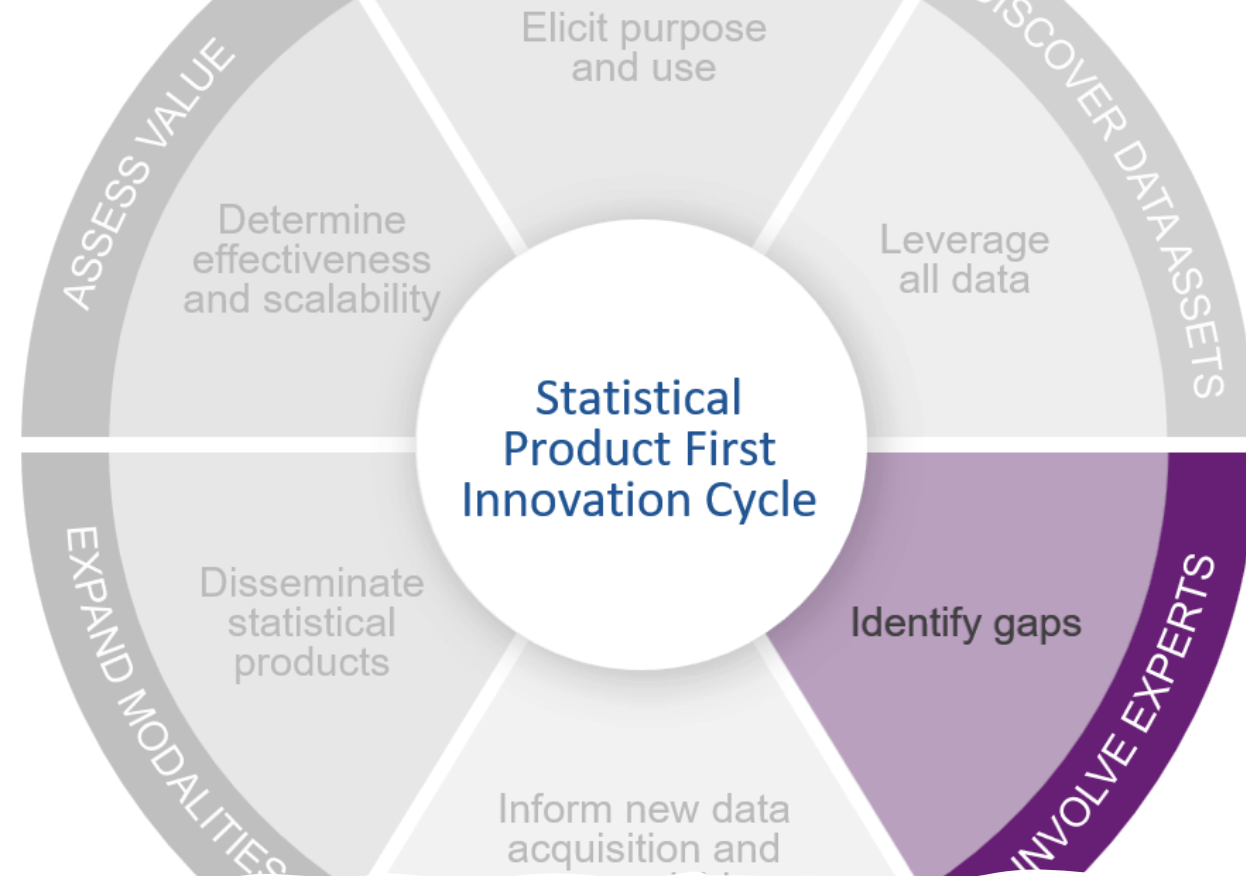
Potential Data Source	Data type	Puzzle piece
Household Trends and Outlook Pulse Survey (HTOPS)	Survey	Timely data on hardship: inability to pay bills, rental payments, etc.
American Community Survey (ACS)	Survey	Large survey, granular stats, numerous topics including veterans
US Veterans Data	Administrative records	Detailed info on veterans
Veteran Employment Outcomes (VEO)	Stat product	Earnings and employment for veterans
HUD Point-in-Time (PIT) Count	Aggregated data	Geographic stats on homelessness
Community Resilience Estimates (CRE)	Stat product	Vulnerability index
And more!	All	Many

Discover Data

Leverage all data

What data can help answer the question?

- Integrate survey, commercial, and administrative data
- Leverage the Census Data Linkage Infrastructure
- Combine complementary data sources to identify risk

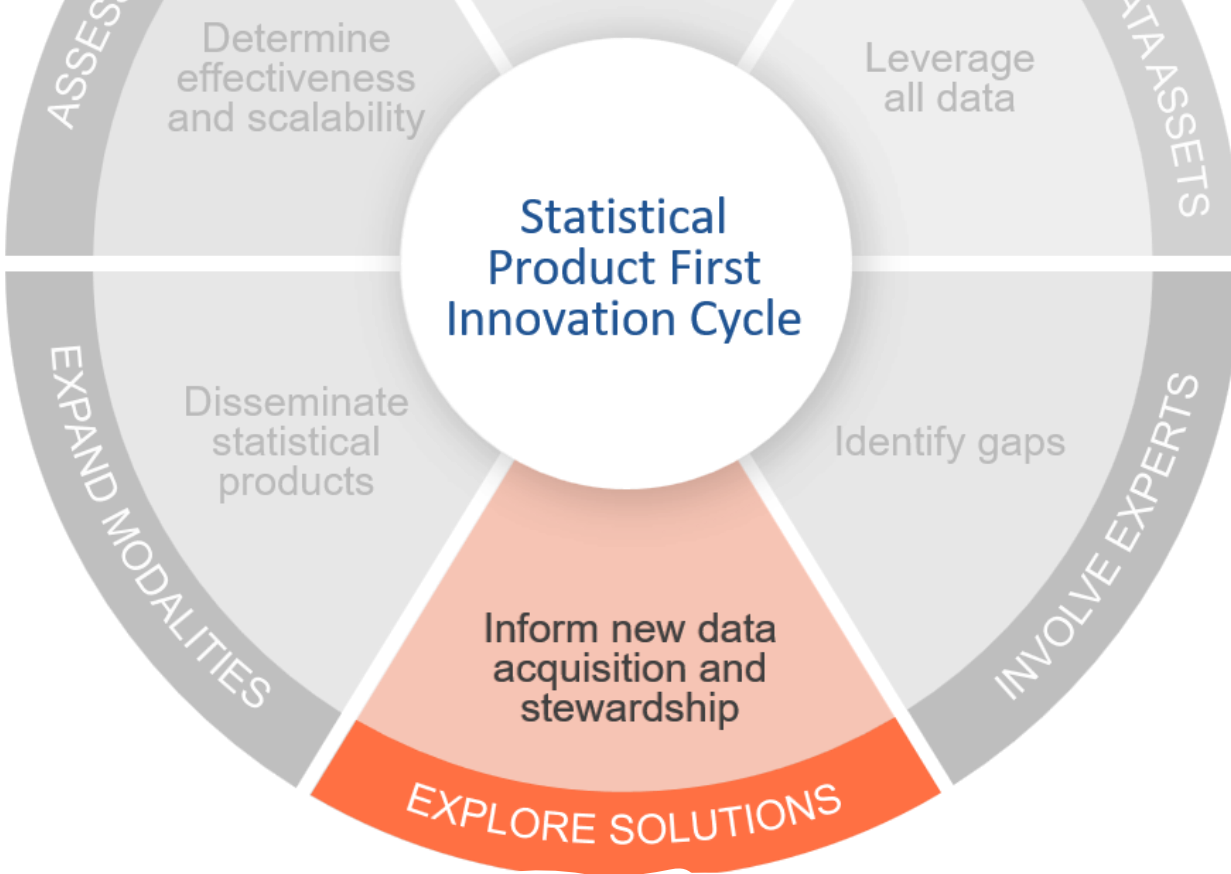


Involve Experts

Identify gaps

Who can help us understand the problem?

- Engage experts across government, academia, and nonprofit organizations
- Refine how risk is defined, measured, and interpreted
- Ensure the statistics reflect real-world decision contexts

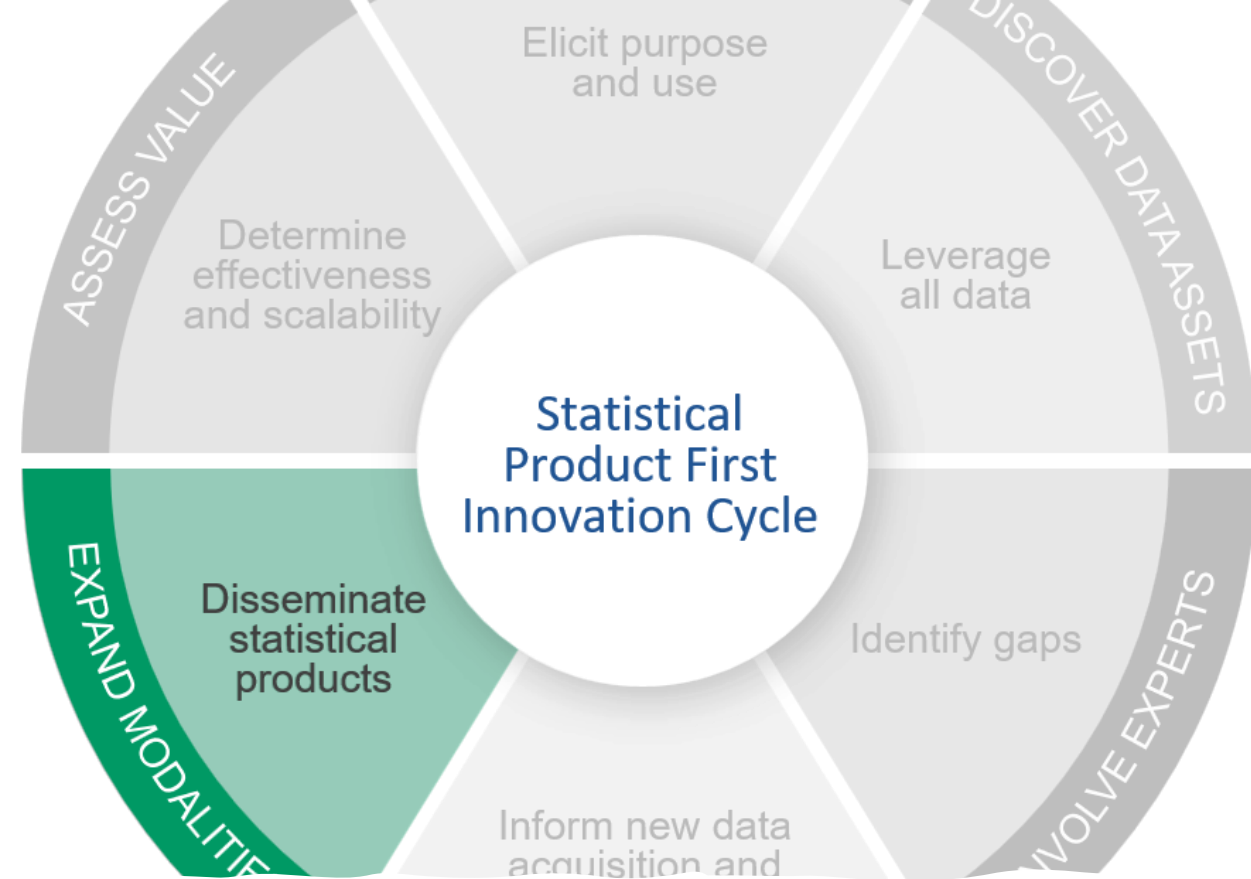


Explore Solutions

Inform data acquisition and stewardship

How should the statistical product be designed?

- Integrate multiple data sources with model-based methods
- Develop indicators of elevated risk—not just outcomes
- Enable earlier intervention through forward-looking statistics

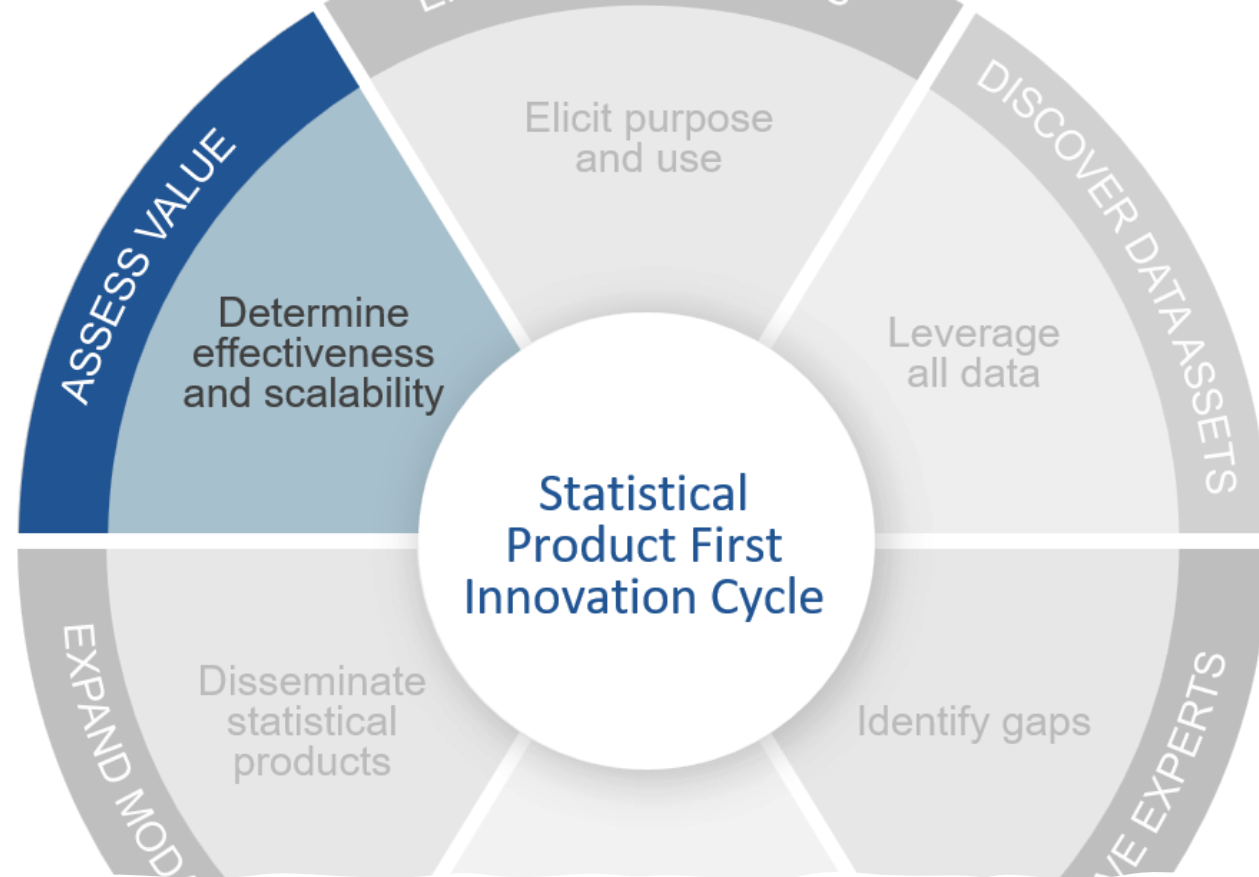


Expand Modalities

Disseminate statistical products

How should the statistics be delivered?

- Present results in formats tailored to stakeholder needs
- Support timely interpretation and action
- Enable practical use in programs, policy, and planning



Assess Value

Determine effectiveness and scalability

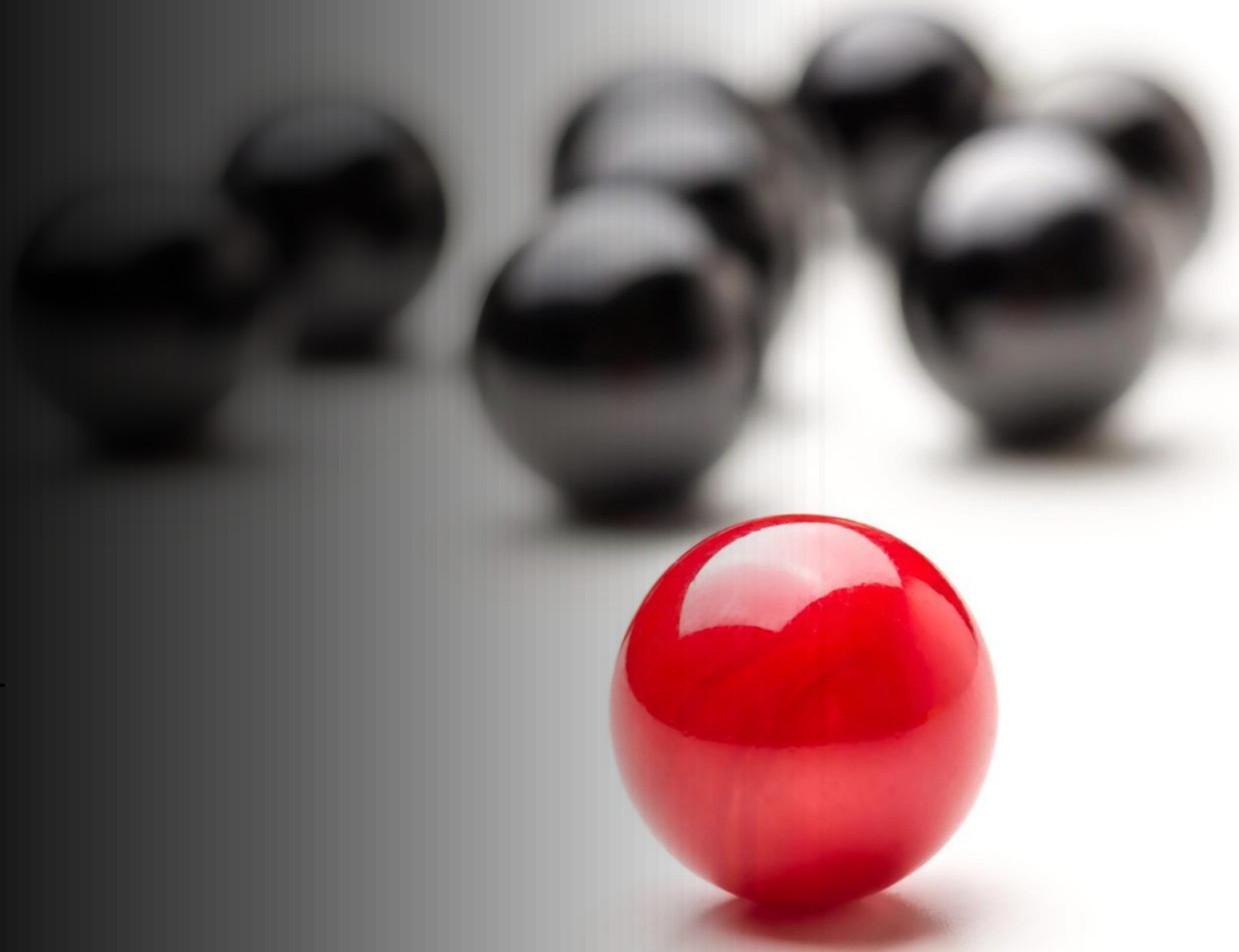
Did the statistical product achieve its purpose?

- Evaluate whether the statistics support earlier intervention
- Refine the product through iterative feedback
- Scale what works; improve what doesn't

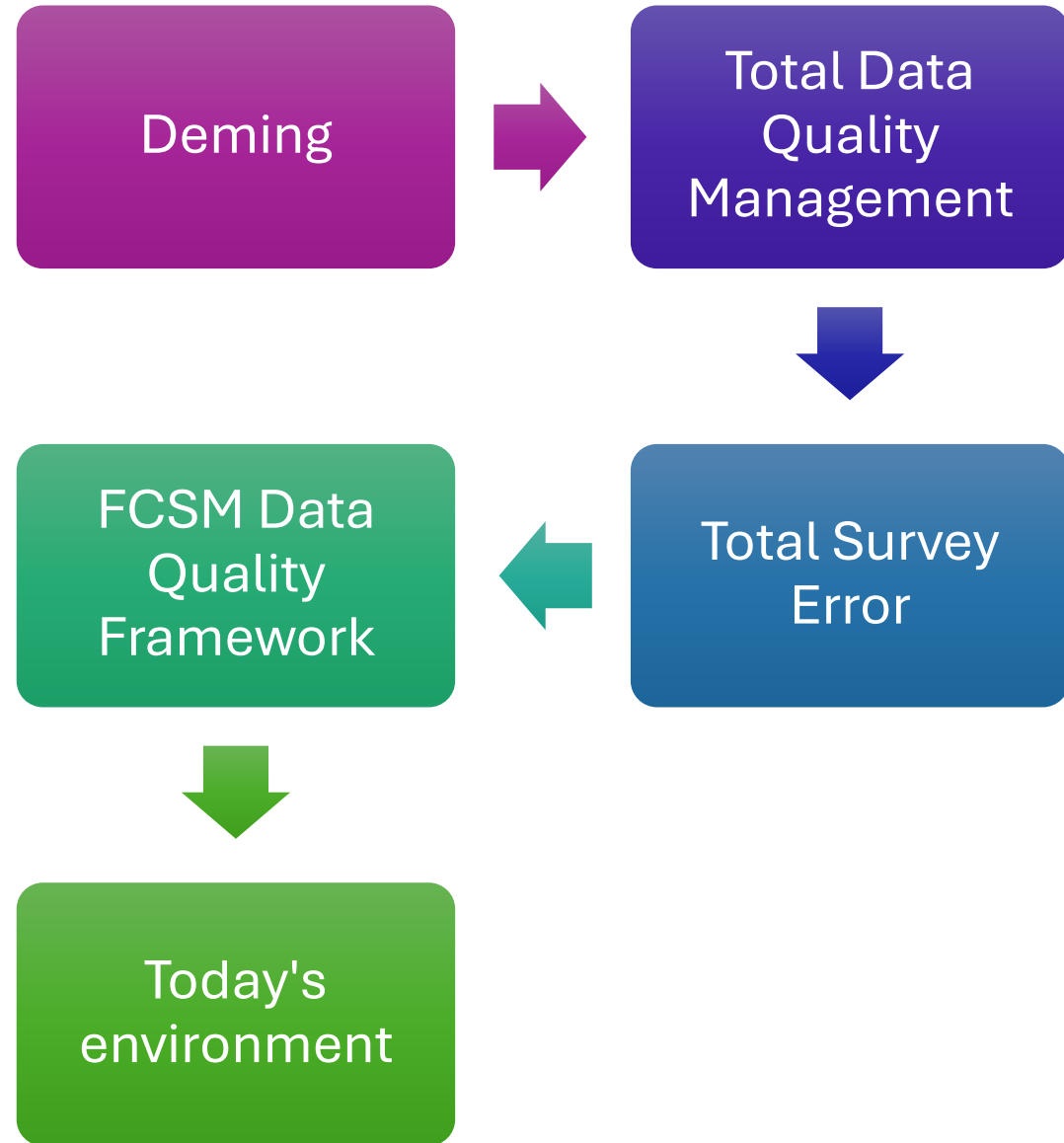


If we've fundamentally
changed how we develop
and produce statistics...

Then why would we expect
the way we think about
quality to remain
unchanged?



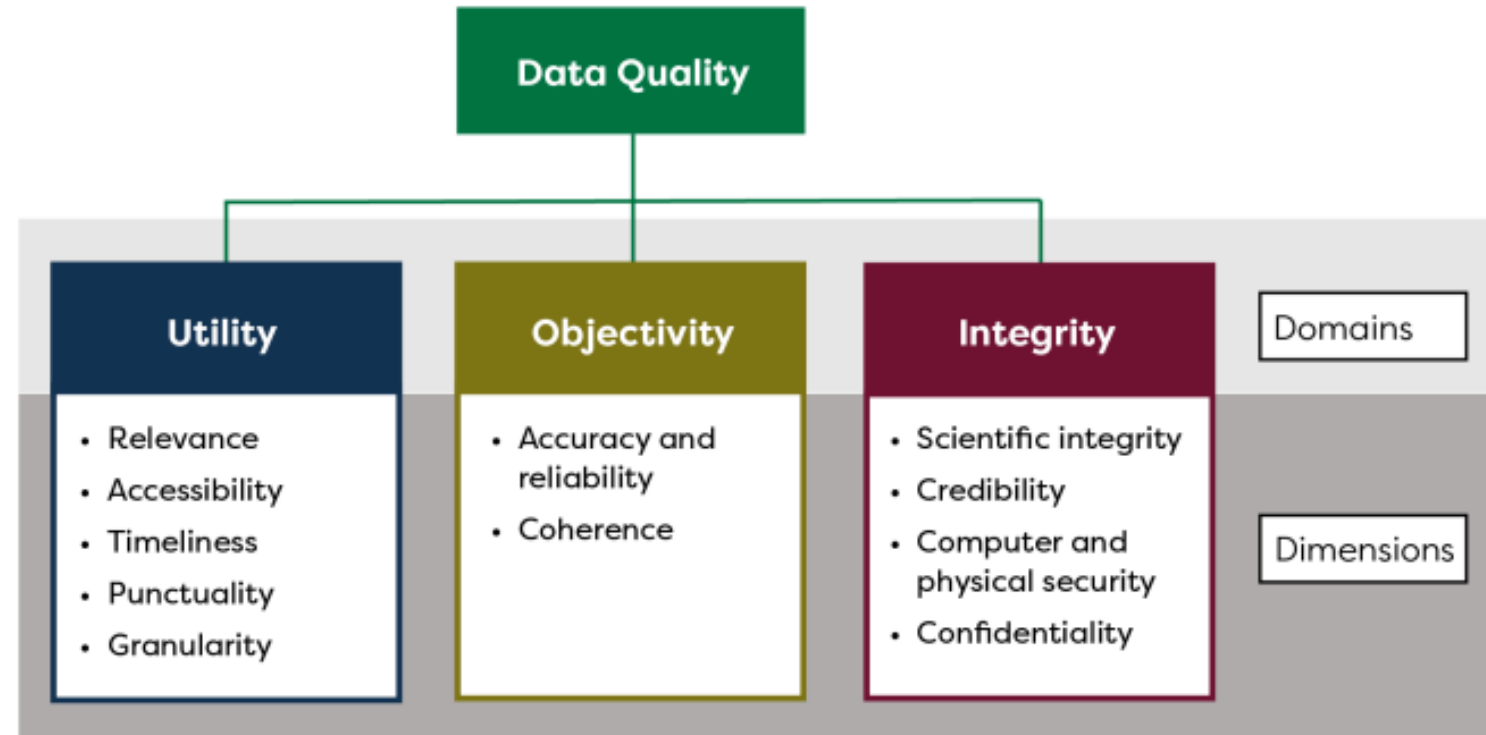
The Evolution of Quality



Data Quality Today

- Traditionally data collection is designed to minimize bias and maximize information content and quality is verified after data is collected.
- This tradition is being challenged: diverse uses, iterative processes for building statistical products, and products based on blended data through massive data repurposing.

2020 FCSM took on the challenge and proposed a data quality framework to support statistical products.



Building Quality In

Transition from
inspection

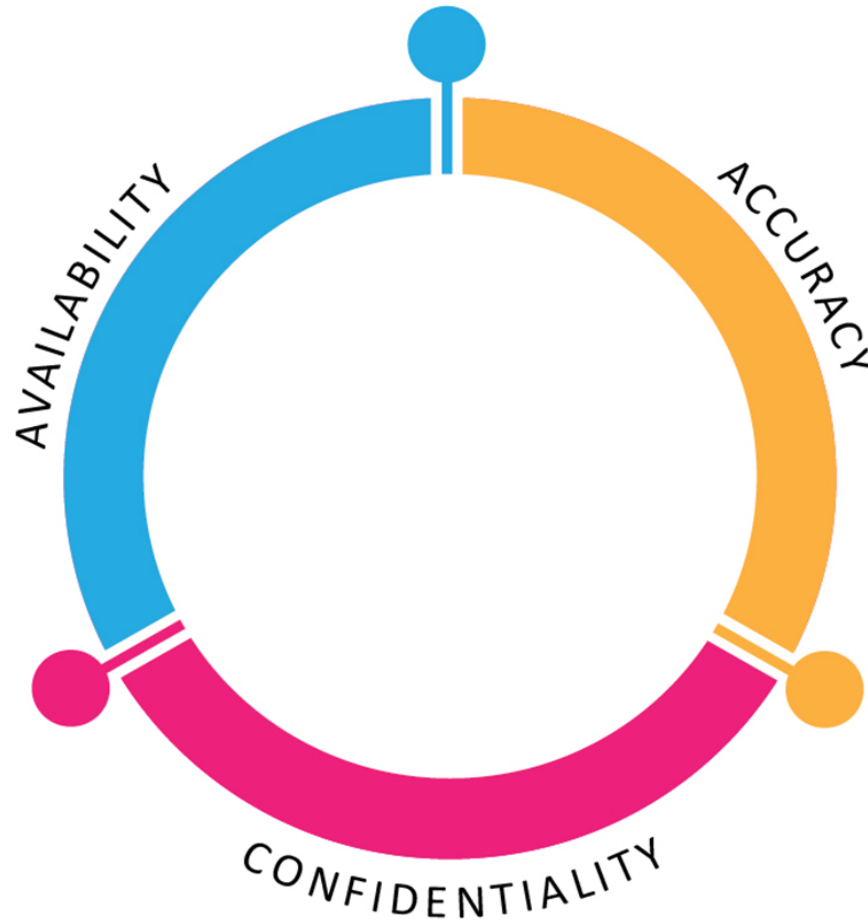


continuous
reasoning



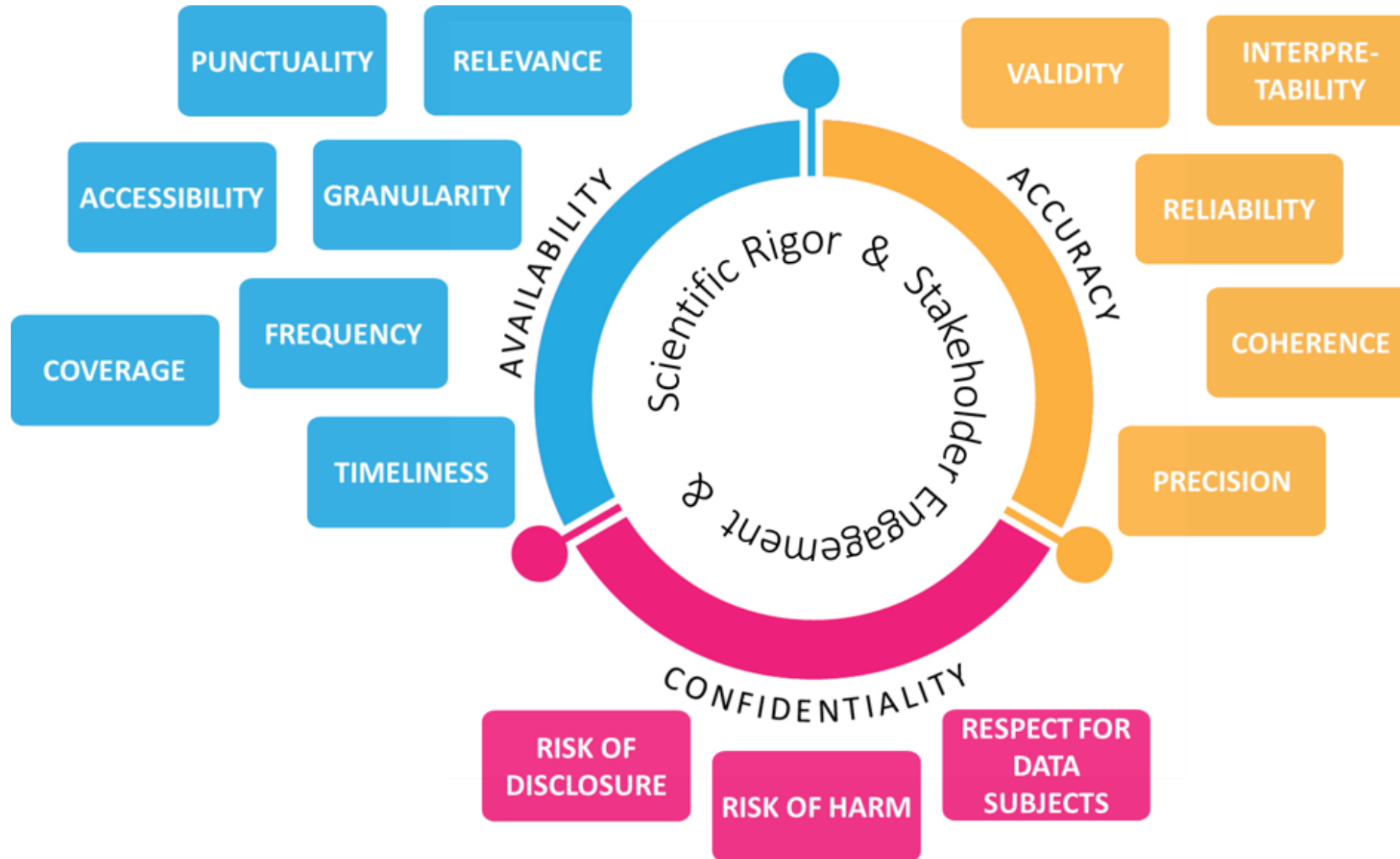
Evolution of an Enterprise Framework

What began as conversations about total survey error and disclosure avoidance evolved into an enterprise model for thinking about quality.

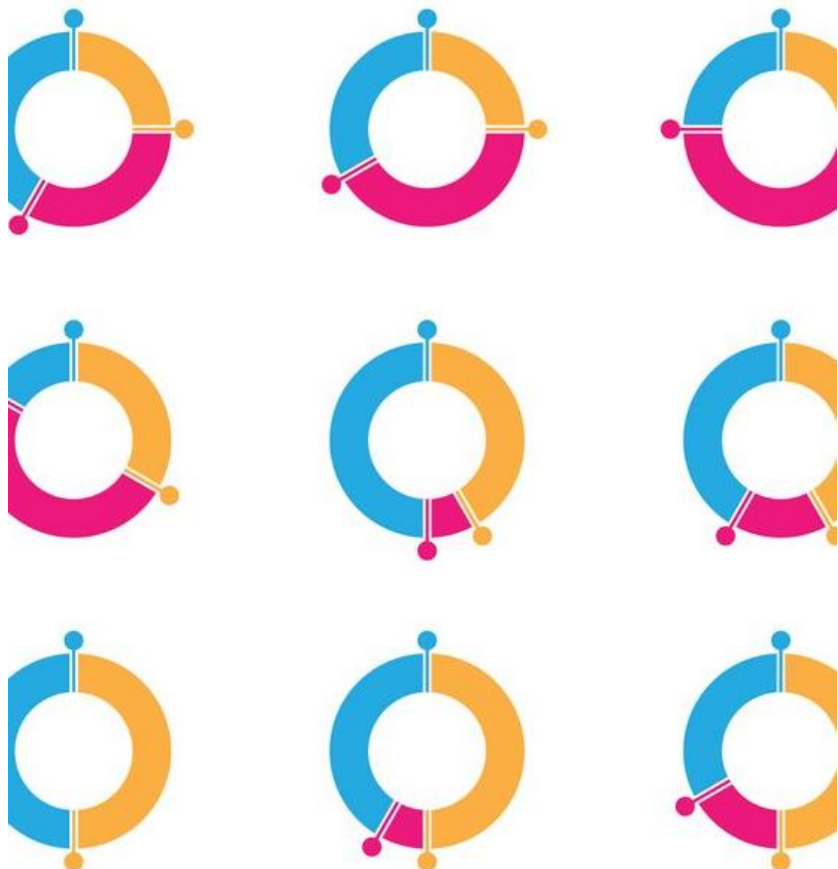


The Triple Tradeoff

Statistical Quality Framework



AVAILABILITY



RELEVANCE

What high value purposes and uses do the statistics need to support? How well do the statistics align with known and anticipated public policy and societal needs?

GRANULARITY

Are the statistics sufficiently detailed and disaggregated to support their intended purpose and use?

FREQUENCY

Are updated statistics released often enough to support their intended purpose and use?

TIMELINESS

How current do the statistics need to be to support their intended purpose and use?

PUNCTUALITY

Are the statistics being released on time? Are user expectations about the release schedule being appropriately managed and met?

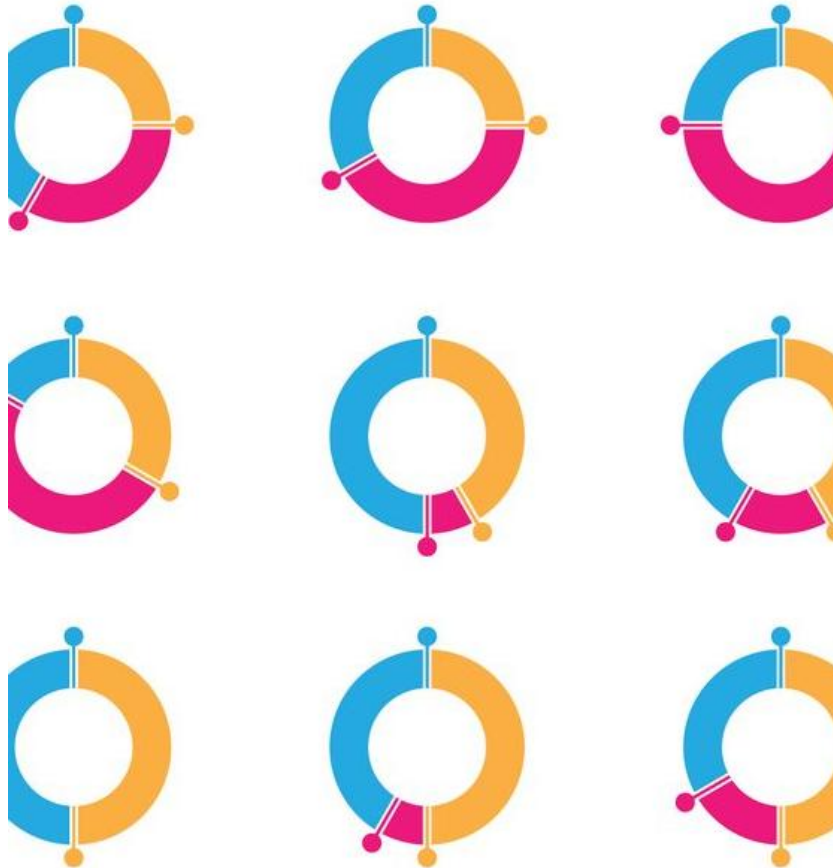
ACCESSIBILITY

Can users of varying levels of data acumen easily obtain the statistics and documentation in understandable formats that facilitate their subsequent analysis and use?

COVERAGE

How well will the statistics support important uses by the broader community? Do they describe and support analysis by the full range of geographies and subpopulations?

ACCURACY



VALIDITY

Are the statistics describing the underlying characteristic of the population that they purport to measure?

RELIABILITY

Are the statistics reproducible?

COHERENCE

Will the statistics align sufficiently with other relevant statistical assets to permit broader purposes and uses?

PRECISION

Is the degree of uncertainty associated with the statistics small enough to support confidence in the analysis or decisions reached using those statistics?

INTERPRETABILITY

Are the statistics readily understandable by their intended users? Do the data users have sufficient information about the potential limitations of the statistics for their intended use?

CONFIDENTIALITY



RISK OF DISCLOSURE

Has the risk of disclosure associated with releasing the statistics been sufficiently mitigated?

RISK OF HARM

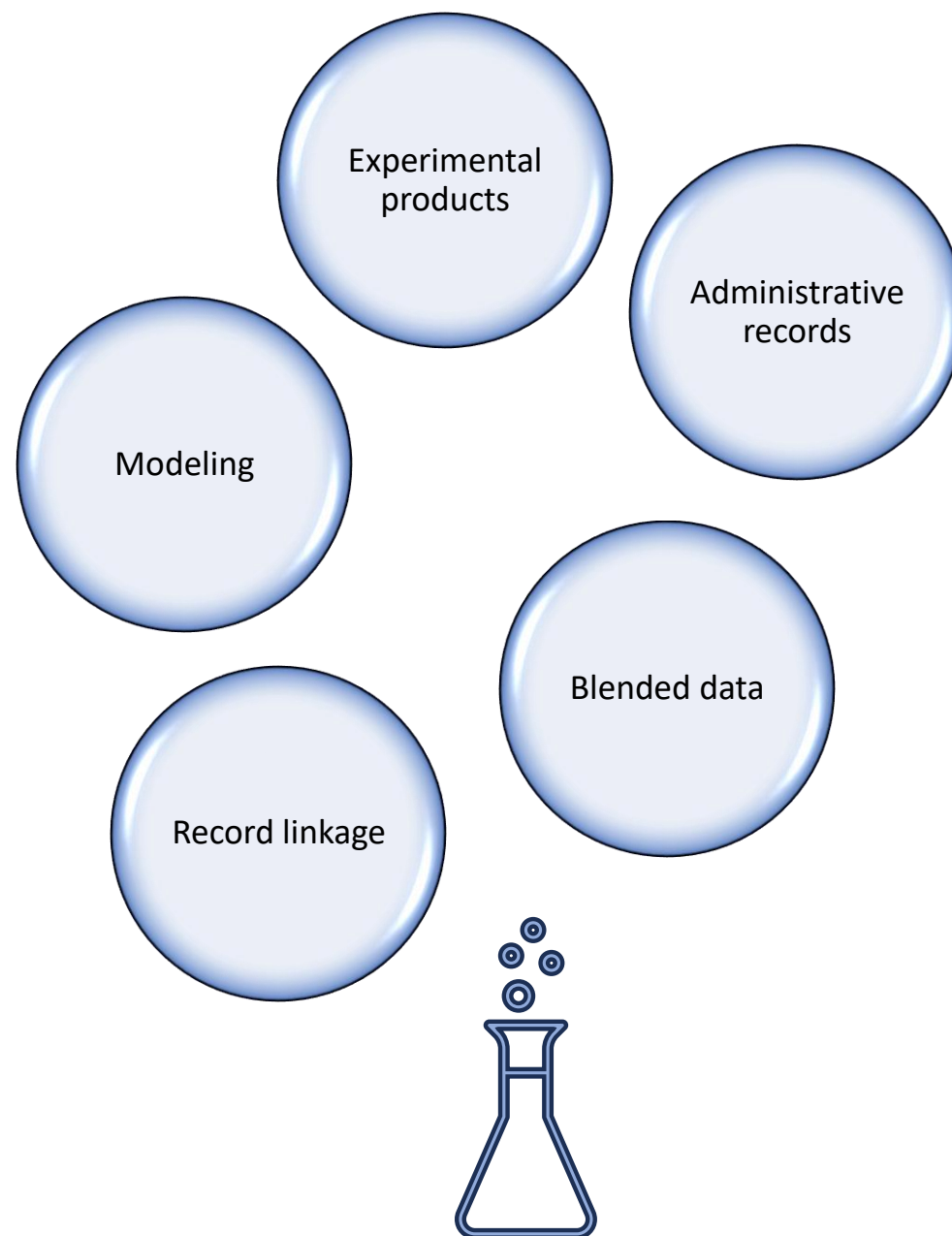
Has the risk of harm to the data subject, if a disclosure were to occur, been sufficiently mitigated?

RESPECT FOR DATA SUBJECTS

Have the statistics been collected, processed, and produced in a manner that respects the privacy and autonomy of the data subject and that promotes trust?

Our Production Reality Has Changed

Our quality
standards were built
for a different era—
a survey-first world



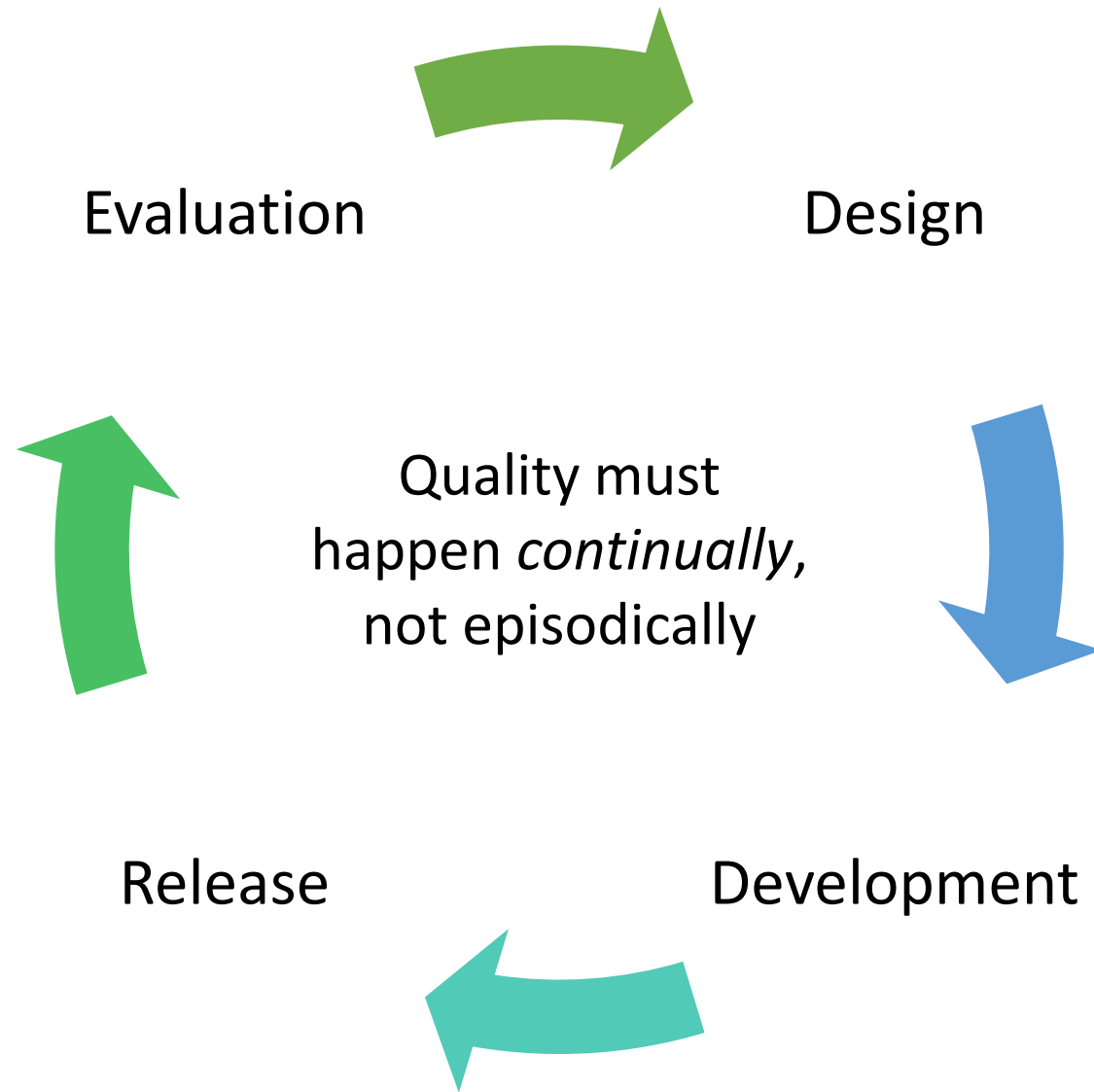
Where Traditional Standards Fall Short

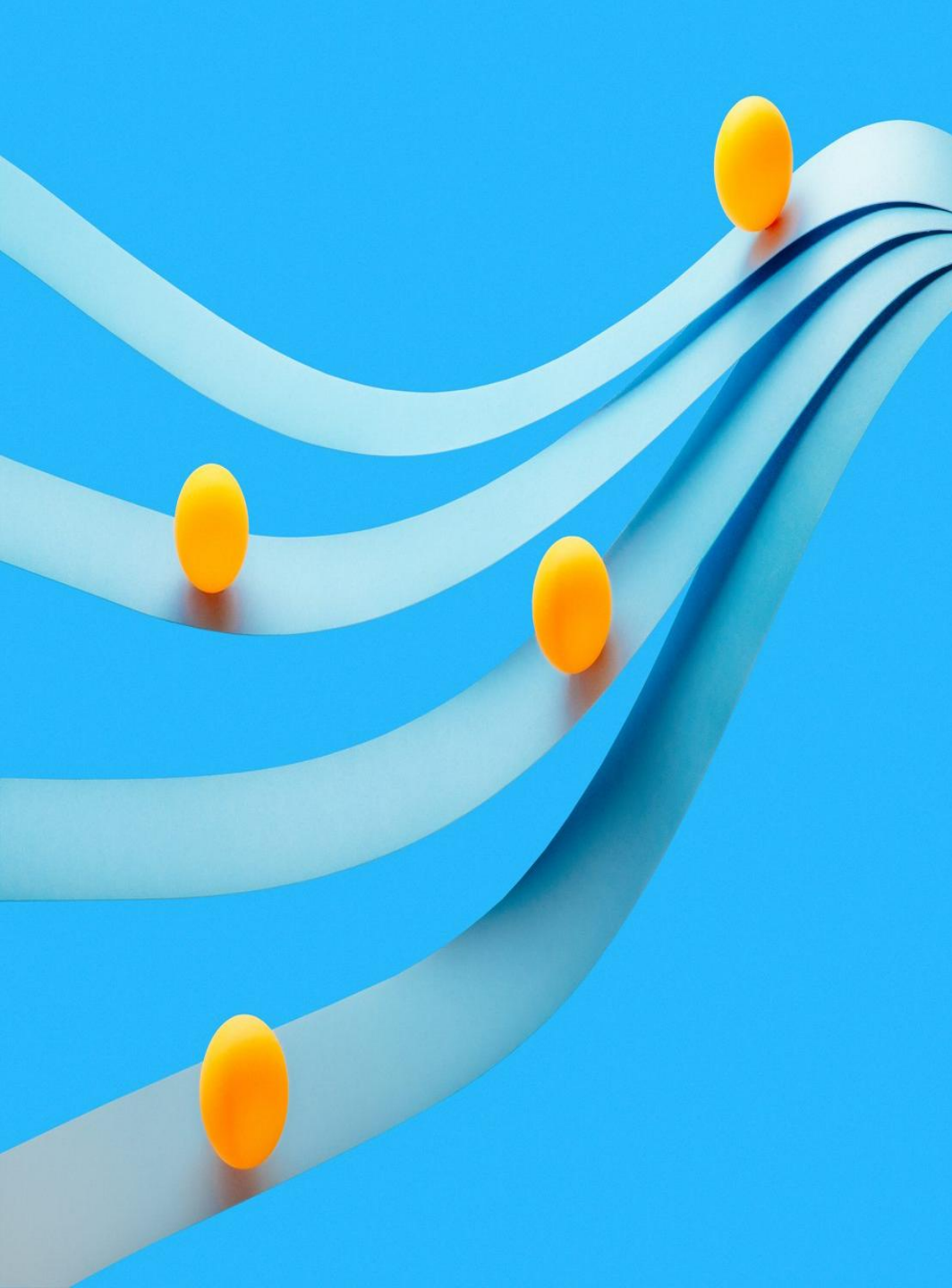
Today's risks go *beyond* bias, variance, and response rates.

- What if we don't control data collection?
- What if risk is misuse, not error?
- What if access = quality?
- What if context is ignored?



Quality Is
Not Only a
Checkpoint
— It's a
Lifecycle
Discipline





A Paradigm Shift

Traditional

Modern

Survey-first

Statistical Product First

Data quality

Statistical quality

Inspect quality

Build quality in

End of production

Throughout the lifecycle

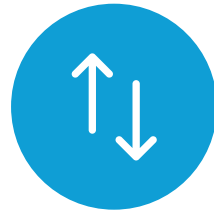
The framework is not a checklist.



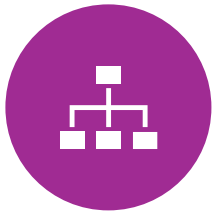
PLANNING



JUDGMENT



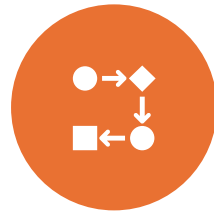
TRADEOFFS



GOVERNANCE



RISK



ITERATION



Pressure testing the
quality framework
with the statistical
product for
“Identifying Risk of
Homelessness for
Veterans”



What decision are we trying to support?

How can we identify indicators associated with elevated risk of homelessness among veterans—**early enough** to support more proactive, targeted intervention?



Quality considerations

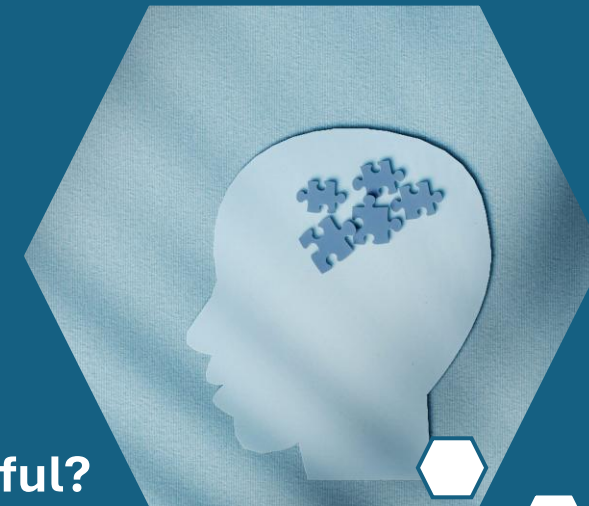
- Relevance

Who needs the statistics—and for what purpose?

Stakeholder	Decision Supported
VA hospitals	Targeted interventions
Federal agencies	Program and policy decisions
State and local stakeholders	Resource allocation

At what geographic level would the statistics be most useful?

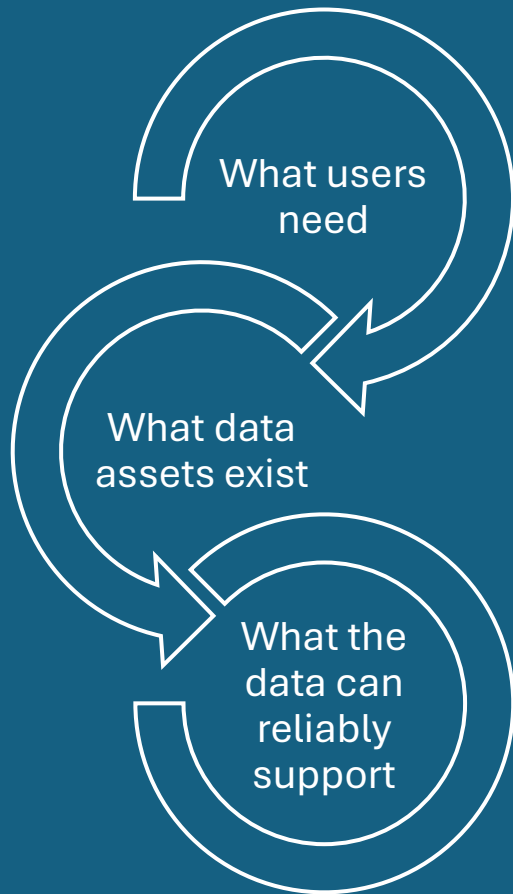
VA catchment areas • States • Urban and rural communities



Quality considerations

- Relevance
- Granularity

What can the available data actually support?



- Are the right veterans represented?
- Can we support the needed geographic detail?
- Can we produce consistent, reliable statistics?



Quality considerations

- Coverage
- Reliability
- Granularity

What happens when every quality goal cannot be maximized?

Greater precision

Finer granularity

Faster delivery

Which matters most for the decision we're trying to support—at this stage of development?



Quality considerations

- Accuracy
- Granularity
- Timeliness

And then the questions change.



Quality considerations

- Relevance
- Coverage
- Accuracy
- Accessibility
- Interpretability

Making quality reasoning explicit.

Many quality considerations were already being addressed—implicitly.

What if we made that reasoning explicit from the beginning?



*Quality considerations integrated
across the statistical product lifecycle*

Quality Considerations Across the Statistical Product Lifecycle

Triple Tradeoff Dimensions →	ACCURACY					AVAILABILITY							CONFIDENTIALITY		
Statistical Product First (SPF) Innovation Cycle Phases ↓	Validity	Reliability	Precision	Interpretability	Coherence	Relevance	Granularity	Frequency	Timeliness	Punctuality	Accessibility	Coverage	Risk of Disclosure	Risk of Harm	Respect for Data Subjects
Engage Data Users															
Discover Data Assets															
Involve Experts															
Explore Solutions															
Expand Modalities															
Assess Value															

The Cultural Shift

Statistical Quality is Both a Process and an Outcome

From

Quality as a checkpoint

Office-specific definitions

Standard-centric
conversations

Siloed decision-making

To

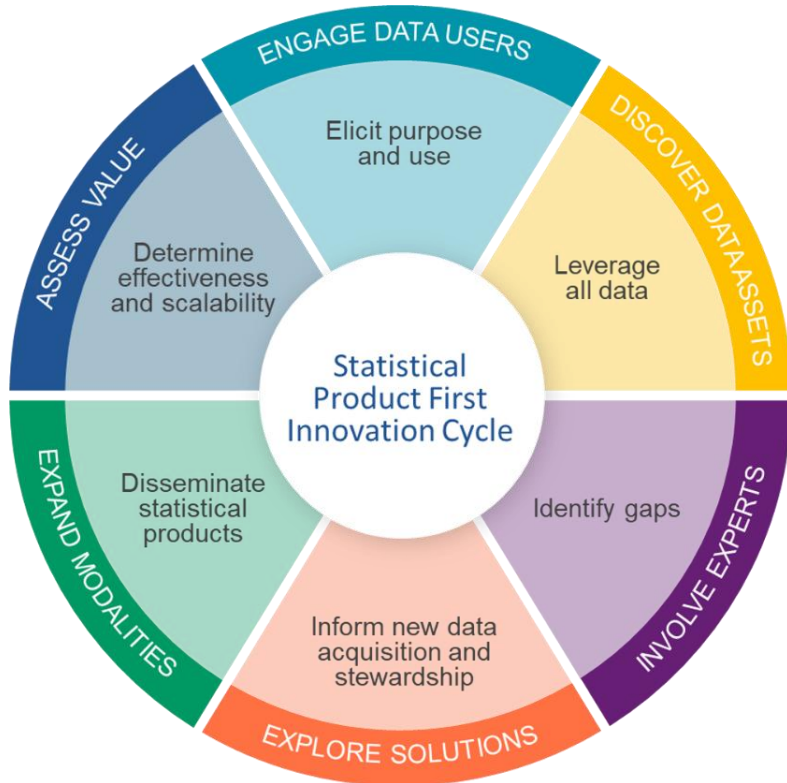
Quality as a lifecycle activity

Enterprise-wide shared logic

Risk-, user-, and phase-aware
reasoning

Transparent, coordinated
decisions—including compliance

Looking Forward



The Statistical Product First (SPF) framework provides a practical way to build quality into statistical product development.

It helps us:

- Start with purpose and use.
- Make quality considerations explicit.
- Continuously improve as products and user needs evolve.

Deming's principle hasn't changed.

The statistical environment has.

Our responsibility is to ensure that the way we build quality into official statistics evolves with it.